SMARTBRIDGE Let's Bridge the Gap

SKILL WALLET A SmartBridge Product

Project Initialization and Planning Phase

Date	30 June 2026
Team ID	
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The OptiCrop: Smart Agricultural Production Optimization Engine is an AI-powered web application that uses Machine Learning to recommend the most suitable crop based on soil nutrients and environmental conditions. The system helps farmers improve crop productivity, optimize resource utilization, and promote sustainable agriculture through accurate, data-driven recommendations.

Project Overview	
Objective	The primary objective of OptiCrop is to develop an intelligent crop recommendation system using Machine Learning that predicts the most suitable crop based on soil nutrients and environmental parameters, enabling farmers to make informed agricultural decisions.
Scope	The project focuses on analyzing soil and environmental data such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall to generate accurate crop recommendations. It also supports researchers and policymakers in analyzing crop-environment relationships for sustainable agricultural planning.
Problem Statement	
Description	Farmers often face difficulties in selecting suitable crops due to changing environmental conditions, lack of scientific guidance, and limited access to data-driven farming solutions. This can result in reduced productivity and inefficient resource utilization.
Impact	Providing intelligent crop recommendations improves agricultural productivity, optimizes the use of soil resources, reduces farming risks, and supports sustainable agricultural practices through data-driven decision-making.
Proposed Solution	
Approach	Develop an AI-powered web application that analyzes soil nutrients and environmental parameters using a trained Machine Learning model to recommend the most suitable crop. The system integrates a Flask backend, MySQL database, and a user-friendly web interface to deliver accurate and real-time crop recommendations.

Key Features	OptiCrop is an AI-powered web application that uses Machine Learning to recommend the most suitable crop based on soil nutrients and environmental conditions. It provides real-time crop predictions through a secure, user-friendly interface, helping farmers improve productivity and support sustainable agriculture.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5 (11th Gen or above) / AMD Ryzen 5, Quad Core
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 256 GB SSD or above
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Visual Studio Code
Data		
Data	Source, size, format	Kaggle Crop Recommendation Dataset, CSV Format